

Tic Tac Toe Game

Aim

To implement a simple **Tic Tac Toe game** using Python.

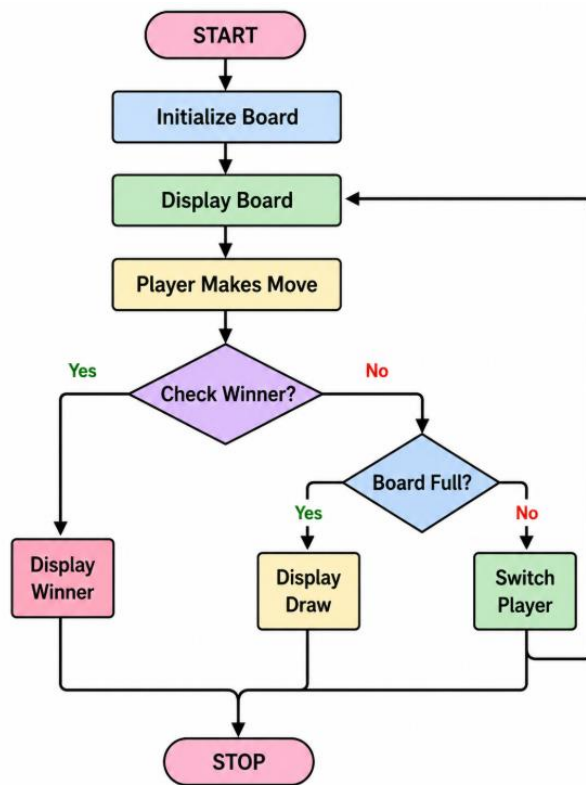
Objectives

- To understand game logic using Python.
- To implement a two-player game.
- To check winning and draw conditions.
- To use functions and loops.

Algorithm

1. Start.
2. Create a 3×3 board.
3. Display the board.
4. Ask the current player to enter a position.
5. Place X or O in the selected position.
6. Check whether the player has won.
7. If there is a winner, display the winner.
8. If all positions are filled, declare a draw.
9. Otherwise, switch the player.
10. Repeat until the game ends.
11. Stop.

Flowchart



ROGRAM

```

print()
print(board[0], "|", board[1], "|", board[2])
print("----")
print(board[3], "|", board[4], "|", board[5])
print("----")
print(board[6], "|", board[7], "|", board[8])

def winner(player):
    combinations = [
        (0,1,2), (3,4,5), (6,7,8),
        (0,3,6), (1,4,7), (2,5,8),
        (0,4,8), (2,4,6)
    ]

    for a, b, c in combinations:
        if board[a] == board[b] == board[c] == player:
            return True
    return False

player = "X"

for turn in range(9):
    display()
    position = int(input("Enter position (1-9): ")) - 1

    if board[position] != " ":
        print("Position already occupied!")
        continue

    board[position] = player

    if winner(player):
        display()
        print("Player", player, "wins!")
        break

    player = "O" if player == "X" else "X"
else:
    display()
    print("Game Draw!")
  
```

OUTPUT

```

+---+
X |  | 
+---+
|  | 
Enter position (1-9): 8
| X | O 
+---+
X |  | 
+---+
| O | 
Enter position (1-9): 6
| X | O 
+---+
X |  | X 
+---+
| O | 
Enter position (1-9): 7
| X | O 
+---+
X |  | X 
+---+
O | O | 
Enter position (1-9): 1
X | X | O 
+---+
X |  | X 
+---+
O | O | 
Enter position (1-9): 5
X | X | O 
+---+
X | O | X 
+---+
O | O | 
Player O wins!
>>>

```

Result

The Tic Tac Toe game was successfully implemented using Python.